

Linear Algebra Exam 3 Study Guide

1 Section 5.2: Inner Product Spaces

- An **inner product** (denoted as $\langle \vec{u}, \vec{v} \rangle$) is literally just any operation that satisfies the following axioms (let $\vec{u}, \vec{v}$, & $\vec{w}$ be in a vector space V , and $c \in \mathbb{R}$):

1. $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$
2. $\langle \vec{u}, \vec{v} + \vec{w} \rangle = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle$
3. $c\langle \vec{u}, \vec{v} \rangle = \langle c\vec{u}, \vec{v} \rangle = \langle \vec{u}, c\vec{v} \rangle$
4. $\langle \vec{u}, \vec{u} \rangle > 0$ & $\langle \vec{u}, \vec{u} \rangle = 0$ iff $\vec{u} = 0$

Note: The dot product is an example of an inner product (it is also called the Euclidean inner product)

- Note: An inner product can also be done on matrices, and even definite integrals
- **Inner product space:** a vector space with an inner product defined
- **Length of a vector:** $\|\vec{u}\| = \sqrt{\langle \vec{u}, \vec{u} \rangle}$
- **Distance between 2 vectors:** $d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|$
- **The angle between 2 non-zero vectors:**

$$\cos(\theta) = \frac{\langle \vec{u}, \vec{v} \rangle}{\|\vec{u}\| \|\vec{v}\|}, \text{ \& } 0 \leq \theta \leq \pi$$

- **Orthogonal:** $\langle \vec{u}, \vec{v} \rangle = 0$
- **Unit vector eq:** $\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$ (or to normalize)
- **Orthogonal Projections:** a vector broken down into 2 parts
 - $\vec{u} = \hat{u} + \vec{z}$
 - * $\vec{u}$: a vector
 - * $\hat{u}$: the projection of $\vec{u}$ on another vector $\vec{v}$. Think of it as the shadow that $\vec{u}$ casts on all scalar multiples of $\vec{v}$
 - * $\vec{z}$: The vector orthogonal to $\vec{v}$
 - $\hat{u} = \text{proj}_{\vec{v}} \vec{u} = \frac{\langle \vec{u}, \vec{v} \rangle}{\langle \vec{v}, \vec{v} \rangle} \vec{v}$
 - $\vec{z} = \vec{u} - \hat{u}$

2 Section 5.3: Orthogonal Basis - Gram-Schmidt

- Vector spaces can have many basis, like how $\mathbb{R}^3$ has the basis $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ (the standard basis for $\mathbb{R}^3$). Standard basis have some important characteristics, like how all the vectors in it are mutually orthogonal

$$(1, 0, 0) \cdot (0, 1, 0) = 0$$

$$(1, 0, 0) \cdot (0, 0, 1) = 0$$

$$(0, 1, 0) \cdot (0, 0, 1) = 0$$

- **Orthogonal set:** $S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} \in \mathbb{R}^n$ such that $\langle \vec{u}_i, \vec{u}_j \rangle = 0 \forall i \neq j$. Basically, all vectors are orthogonal to each other
- **Othonormal set:** $S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} \in \mathbb{R}^n$ such that: $\langle \vec{u}_i, \vec{u}_j \rangle = 0 \forall i \neq j$, and every vector is a unit vector.
- **Gram-Schmidt Orthonormalization Algorithm:** Algorithm to find orthonormal bases, where you start with a basis, you turn it into an orthonormal basis, then normalize each vector.

1. Let $B = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ be a basis for an inner product space V

2. Let $B' = \{\vec{w}_1, \vec{w}_2, \dots, \vec{w}_n\}$, where

$$\vec{w}_1 = \vec{v}_1$$

$$\vec{w}_2 = \vec{v}_2 - \frac{\langle \vec{v}_2, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} \vec{w}_1$$

$$\vec{w}_3 = \vec{v}_3 - \frac{\langle \vec{v}_3, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} \vec{w}_1 - \frac{\langle \vec{v}_3, \vec{w}_2 \rangle}{\langle \vec{w}_2, \vec{w}_2 \rangle} \vec{w}_2$$

$$\vdots$$

$$\vec{w}_n = \vec{v}_n - \frac{\langle \vec{v}_n, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} \vec{w}_1 - \frac{\langle \vec{v}_n, \vec{w}_2 \rangle}{\langle \vec{w}_2, \vec{w}_2 \rangle} \vec{w}_2 - \dots - \frac{\langle \vec{v}_n, \vec{w}_{n-1} \rangle}{\langle \vec{w}_{n-1}, \vec{w}_{n-1} \rangle} \vec{w}_{n-1}$$

Then B' is an orthonormal basis for V

3. Let $\vec{u}_i = \frac{\vec{w}_i}{\|\vec{w}_i\|}$. Then $B'' = \{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n\}$ is an orthonormal basis for V

Also, $\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} = \text{span}\{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n\}$ for $k = 1, 2, \dots, n$

3 Section 6.1: Introductions to Linear Transformations

- Get back to this